

swtpcsim — An interrupt-driven clock on the MP-T timer

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The SWTPC **MP-T** is an interrupt timer board: a 6820 PIA in an SS-30 slot, with a Mostek MK5009 time base behind its B side. A program picks a rate, from 1 μ s to one hour, and the board interrupts the 6800 once a period. This example runs **INTCLK**, the clock program SWTPC published with the board. It asks for the time, then keeps it on the console, one tick per interrupt.

```
cd examples/mpt
swtpcsim mpt.toml
```

The machine file loads `INTCLK.S19` into memory and starts SWTBUG, so you land at its `$` prompt. Type `G` to run the program:

```
$ G
SWTPC 6800 COMPUTER SYSTEM TIME:
HH:MM.SS
```

Wait for the `HH:MM.SS` line, then type the time: all eight characters, as `HH:MM.SS` (`12:59.58` , `09:05.00`). The clock is in 12-hour format. **INTCLK** takes no Return. After the eighth character it starts the timer, and the time starts to move, one second at a time:

```
12:59.58
12:59.59
1:00.00
1:00.01
```

In a terminal, each new time overwrites the last on the same line. `Ctrl-E` returns to the `swtpcsim>` prompt.

Three things you will see are what the real machine does:

- **The first tick comes at once.** **INTCLK** sets the MP-T up in an order that lets the MK5009 run at 1 μ s for a few instructions before the program holds it. That sets the board's interrupt flag, and the program never clears the flag before it enables interrupts. So the time you type appears one second on, straight away. After that, the ticks are a second apart.
- **The hour has no leading zero.** After `12` , **INTCLK** writes a space: `1:00.00` .
- **Characters you type early are lost.** SWTBUG's output routine reads the console's receive register before every character it prints. Anything you type before `HH:MM.SS` has finished printing is

dropped, so wait for it.

The machine runs at a real 1 MHz (`clock_hz = 1000000` in `mpt.toml`), so a second on the console is a second on the wall. At full host speed, the MK5009, which counts the machine's own microseconds, would run the clock many times too fast.

The files

File	What it is
<code>mpt.toml</code>	The SWTPC 6800 with SWTBUG, the MP-S console, 40K of RAM, and the MP-T in SS-30 slot 4 (<code>8010 – 8013</code>).
<code>INTCLK.ASM</code>	The program's source.
<code>INTCLK.S19</code>	The program as Motorola S-records, which <code>mpt.toml</code> loads.

Both files are corrected from the published copies. In SWTPC's listing, the two words at `A000` and `A002` are the wrong way round. SWTBUG's interrupt handler jumps through the word at `A000` , and INTCLK reads the MP-T's address from `A002` . So `A000` must hold the interrupt routine (`013D`) and `A002` the board (`8010`). The published S-records put them the other way round, and the first interrupt jumps into the MP-T's registers. The header comment's "port #5" is also corrected: with four addresses to a slot, `8010` is slot 4. Nothing else is changed. The source's opening comment says the same.

How the MP-T is programmed

INTCLK does what any MP-T program does, at the board's base address `8010` :

Write	To	Why
<code>FF</code>	<code>8012</code> (DDRB)	All eight B lines become outputs.
<code>3D</code>	<code>8013</code> (CRB)	Select the data register. CB1 interrupts on its falling edge.
<code>80</code>	<code>8012</code> (PRB)	PB7 high holds the MK5009 at zero.
<code>06</code>	<code>8012</code> (PRB)	PB7 low starts it; <code>6</code> selects one second.

Its interrupt routine reads `8012` , which clears the flag, then adds a second. The board and its rate table are in the manual's `mpt` section.